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COURSE CERTIFICATE

Feb 28, 2026

Shatakshi Tripathi

has successfully completed

NVIDIA: Fundamentals of Deep Learning

an online course authorized by Whizlabs and offered through Coursera

Krishna Srinivasan,
Co-Founder & CEO

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Introduction to Deep Learning & Neural Networks - Knowledge check

Review Learning Objectives

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Assignment details

Submitted	Attempts
Feb 28, 8:56 PM +09	Unlimited

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Your grade

To pass you need at least 60%. We keep your highest score.

100%

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Introduction to Deep Learning & Neural Networks - Knowledge check

Practice Assignment • 15 min

1 point

1. Which layer in a DNN is responsible for receiving the raw input data?

- (●) Input Layer
- ( ) Hidden Layer
- ( ) It determines the accuracy of the model.
- ( ) It has no significant role in machine learning

1 point

2. Which type of Deep Neural Network is best suited for processing images and videos?

- ( ) Multi-Layer Perceptron (MLP)
- (●) Convolutional Neural Network (CNN)
- ( ) Recurrent Neural Network (RNN)
- ( ) Generative Adversarial Network (GAN)

1 point

3. What role do Weights play in an artificial neuron?

- ( ) Introduce non-linearity
- (●) Determine the importance of each input
- ( ) C: Provide a constant offset
- ( ) Receive the initial data

1 point

3. What role do Weights play in an artificial neuron?

- ○ Introduce non-linearity
- ◉ Determine the importance of each input
- ○ C: Provide a constant offset
- ○ Receive the initial data

1 point

4. What is the first step in the computation process of an artificial neuron?

- ○ Apply the activation function
- ◉ Calculate the weighted sum
- ○ Transmit the output signal
- ○ Adjust the bias

1 point

5. What is the initial step in the Gradient Descent algorithm?

- ○ Calculate the gradient of the loss function
- ○ Update the model parameters
- ◉ Initialize the model parameters with random values
- ○ Repeat steps until convergence

Foundations of Deep Learning - Assessment

Review Learning Objectives

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Foundations of Deep Learning - Assessment

Graded Assignment • 20 min

1 point

1. What is the mathematical formula for the ReLU activation function?

- (●) f(x) = max(0, x)
- ( ) f(x) = 1 / (1 + e^-x)
- ( ) f(x) = tanh(x)
- ( ) f(x) = x

1 point

2. What is the purpose of applying an activation function in a neuron?

- ( ) To normalize the input values
- (●) To introduce non-linearity into the model
- ( ) To calculate the weighted sum of inputs
- ( ) To produce the final prediction

1 point

3. In the equation Z = W * X + b, what does Z represent?

- ( ) The weight matrix
- ( ) The input matrix
- ( ) The bias vector
- (●) The matrix of weighted sums for all neurons in a layer

Foundations of Deep Learning - Assessment
Graded Assignment • 20 min

1 point

4. What is the primary goal of backpropagation in neural networks?

- ○ To initialize the model's parameters
- ○ To make predictions on new data
- ◉ To minimize the overall error and improve model accuracy
- ○ To introduce non-linearity into the model

1 point

5. Which step involves feeding the input data through the network to generate a prediction?

- ◉ Forward Pass
- ○ Loss Calculation
- ○ Backward Pass
- ○ Weight Initialization

1 point

6. Given a categorical feature with values ['red', 'green', 'blue'], what would be the one-hot encoded representation of 'green'?

- ○ [1, 0, 0]
- ◉ [0, 1, 0]
- ○ [0, 0, 1]
- ○ [1, 1, 0]

Deep Learning & Transfer Learning Techniques - Knowledge check

Review Learning Objectives

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Deep Learning & Transfer Learning Techniques - Knowledge check

Practice Assignment • 15 min

1 point

1. What type of data are Convolutional Neural Networks (CNNs) primarily designed to process?

- ○ Sequential data, such as text or time series
- ○ Tabular data with structured features
- ◉ Grid-like data, such as images and video
- ○ Audio data

1 point

2. What is the primary purpose of Pooling Layers in a CNN?

- ○ To increase the spatial dimensions of the data
- ○ To introduce non-linearity into the model
- ◉ To reduce the spatial dimensions of the data by downsampling
- ○ To generate the final output predictions

1 point

3. What is the core concept behind transfer learning?

- ○ Training a model from scratch on a small dataset.
- ◉ Leveraging knowledge from a pre-trained model on a new but related task.
- ○ Creating a completely new neural network architecture for every task.
- ○ Only using labeled data for training.

1 point

4. In which scenario is transfer learning most likely to be beneficial?

- ◉ You have abundant labeled data for your specific task.
- ○ The pre-trained model was trained on a task completely unrelated to your target task.
- ○ You have ample computational resources and a large dataset for your new task.
- ○ You have a small dataset for your specific task and limited computational resources.

Advanced Deep Learning Techniques - Assessment

Review Learning Objectives

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Advanced Deep Learning Techniques - Assessment
Graded Assignment • 15 min

1. When loading the VGG16 model, what does setting include_top=False signify?

- (●) It excludes the final fully connected classification layers of the model
- ( ) It excludes the convolutional base of the model
- ( ) It loads the model without pre-trained weights
- ( ) It disables transfer learning

1 point

2. What is the purpose of freezing layers in the VGG16 model during transfer learning?

- (●) To prevent the pre-trained weights from being updated during training
- ( ) To slightly speed up the training process
- ( ) To ensure the model learns only from the new data
- ( ) To reduce the model's complexity

1 point

3. Which activation function maps the input to a range between 0 and 1 and is historically popular but suffers from vanishing gradients?

- (●) Sigmoid
- ( ) Hyperbolic Tangent (tanh)
- ( ) Rectified Linear Unit (ReLU)
- ( ) Linear

3. Which activation function maps the input to a range between 0 and 1 and is historically popular but suffers from vanishing gradients?

- (●) Sigmoid
- ( ) Hyperbolic Tangent (tanh)
- ( ) Rectified Linear Unit (ReLU)
- ( ) Linear

1 point

4. Which activation function is similar to sigmoid but maps the input to a range between -1 and 1?

- ( ) A: Sigmoid
- (●) Hyperbolic Tangent (tanh)
- ( ) Rectified Linear Unit (ReLU)
- ( ) Linear

1 point

5. Which of the following is a common approach in transfer learning?

- ( ) Training a model from scratch with random weights
- ( ) Completely discarding pre-trained models in every training iteration
- ( ) Avoiding the use of neural networks
- (●) Using a pre-trained model as a feature extractor and fine-tuning only specific layers